Assignment 1

Object Oriented Programming

Question #1:

You are designing a class Book for a library system. Each book has a **title**, **author**, **price, Available Quantity,** and **Category** of the book. You also need a function that calculates a **discounted price,** and **Order a Book for purchase** based on a given discount percentage. Make sure all data members should not be accessible for outer programing world. 1 constructor set all data member default values, and another should initialize it to fixed values.

Code:

Book.h

```
#include <iostream>
#include <cstring>
#include <iomanip>
using namespace std;
class Book {
private:
    char* m_title;
    char* m_author;
    char* m_category;
    double m_price;
    int m_quantity;
public:
    Book();
```

```
    Book(const char*, const char*, const char*, double, int);
    ~Book();

    void setTitle(const char*);
    void setAuthor(const char*);
    void setCategory(const char*);
    void setPrice(double);
    void setQuantity(int);

    char* getTitle();
    char* getAuthor();
    char* getCategory();
    double getPrice();
    int getQuantity();

    void orderBook(int);
    double calculateDiscount(int);

    void printBook();
};
```

Book.cpp

```
#include "Book.h"


Book::Book() {
    m_title = nullptr;
    m_author = nullptr;
    m_category = nullptr;
    m_price = 0.0;
    m_quantity = 0;
}


Book::Book(const char* title, const char* author, const char* category, double price, int quantity) {
    m_title = new char[strlen(title) + 1];
    strcpy(m_title, title);


    m_author = new char[strlen(author) + 1];
    strcpy(m_author, author);


    m_category = new char[strlen(category) + 1];
    strcpy(m_category, category);


    m_price = price;
    m_quantity = quantity;
```

```
}

Book::~Book() {
    delete[] m_title;
    delete[] m_author;
    delete[] m_category;
}

void Book::setTitle(const char* title) {
    delete[] m_title;
    m_title = new char[strlen(title) + 1];
    strcpy(m_title, title);
}

void Book::setAuthor(const char* author) {
    delete[] m_author;
    m_author = new char[strlen(author) + 1];
    strcpy(m_author, author);
}

void Book::setCategory(const char* category) {
    delete[] m_category;
    m_category = new char[strlen(category) + 1];
    strcpy(m_category, category);
```

```
}

void Book::setPrice(double price) {
    m_price = price;
}

void Book::setQuantity(int quantity) {
    m_quantity = quantity;
}

char* Book::getTitle() {
    if (m_title == nullptr) return nullptr;
    char* temp = new char[strlen(m_title) + 1];
    strcpy(temp, m_title);
    return temp;
}

char* Book::getAuthor() {
    if (m_author == nullptr) return nullptr;
    char* temp = new char[strlen(m_author) + 1];
    strcpy(temp, m_author);
    return temp;
}
```

```
char* Book::getCategory() {
    if (m_category == nullptr) return nullptr;
    char* temp = new char[strlen(m_category) + 1];
    strcpy(temp, m_category);
    return temp;
}

double Book::getPrice() {
    return m_price;
}

int Book::getQuantity() {
    return m_quantity;
}

double Book::calculateDiscount(int disc) {
    return m_price - (m_price * disc / 100);
}

void Book::orderBook(int quantity) {
    if (quantity > m_quantity) {
        cout << "Not enough stock available! Only " << m_quantity << " books left." << endl;
        return;
    }
```

```
    cout << "Do you want to redeem discount? (y/n): " << endl;
    char choice;
    cin >> choice;

    double total = 0.0;
    if (choice == 'y' || choice == 'Y') {
        int disc;
        cout << "Enter discount percentage: ";
        cin >> disc;
        total = calculateDiscount(disc) * quantity;
    } else {
        total = m_price * quantity;
    }
    m_quantity -= quantity;
    cout << "Order placed successfully!" << endl;
    cout << "Total amount: " << total << endl;
    cout << "Remaining stock: " << m_quantity << endl;
}

void Book::printBook() {
    //This function can also get values without getters
    cout << "Title: " << getTitle() << endl;
    cout << "Author: " << getAuthor() << endl;
```

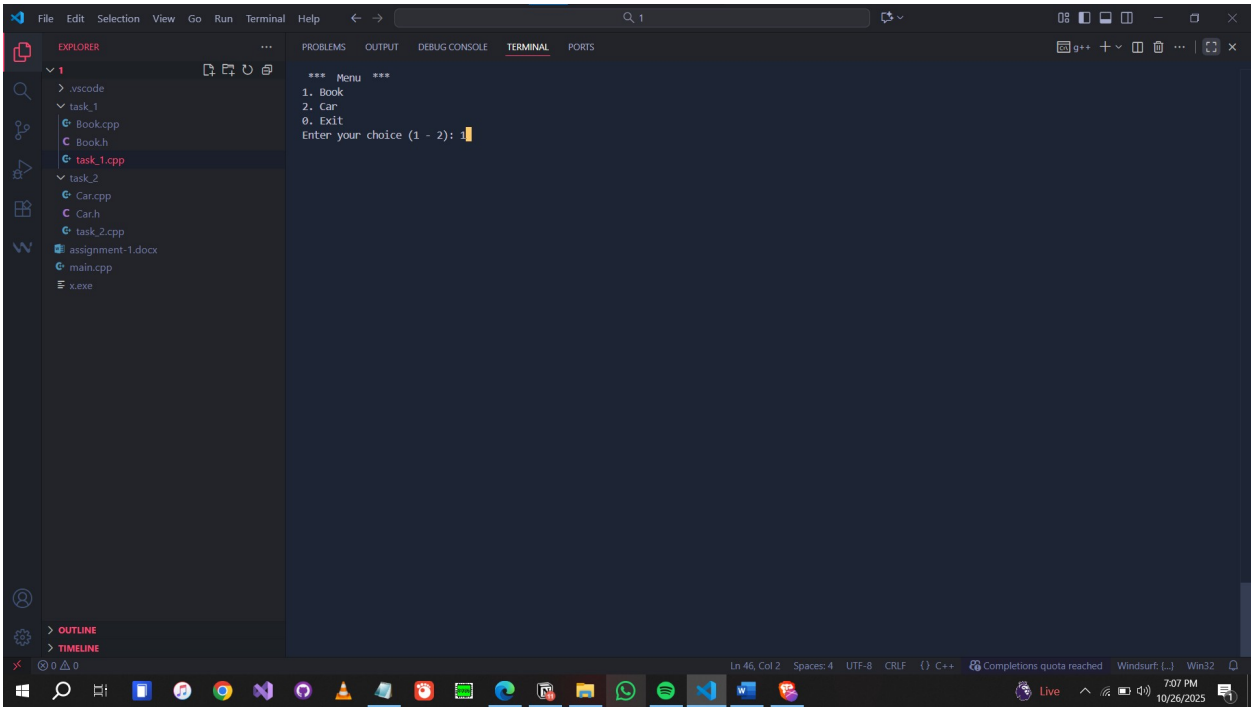
```
    cout << "Category: " << getCategory() << endl;
    cout << "Price: " << fixed << setprecision(3) << getPrice() << endl;
    cout << "Quantity: " << getQuantity() << endl;
}
```

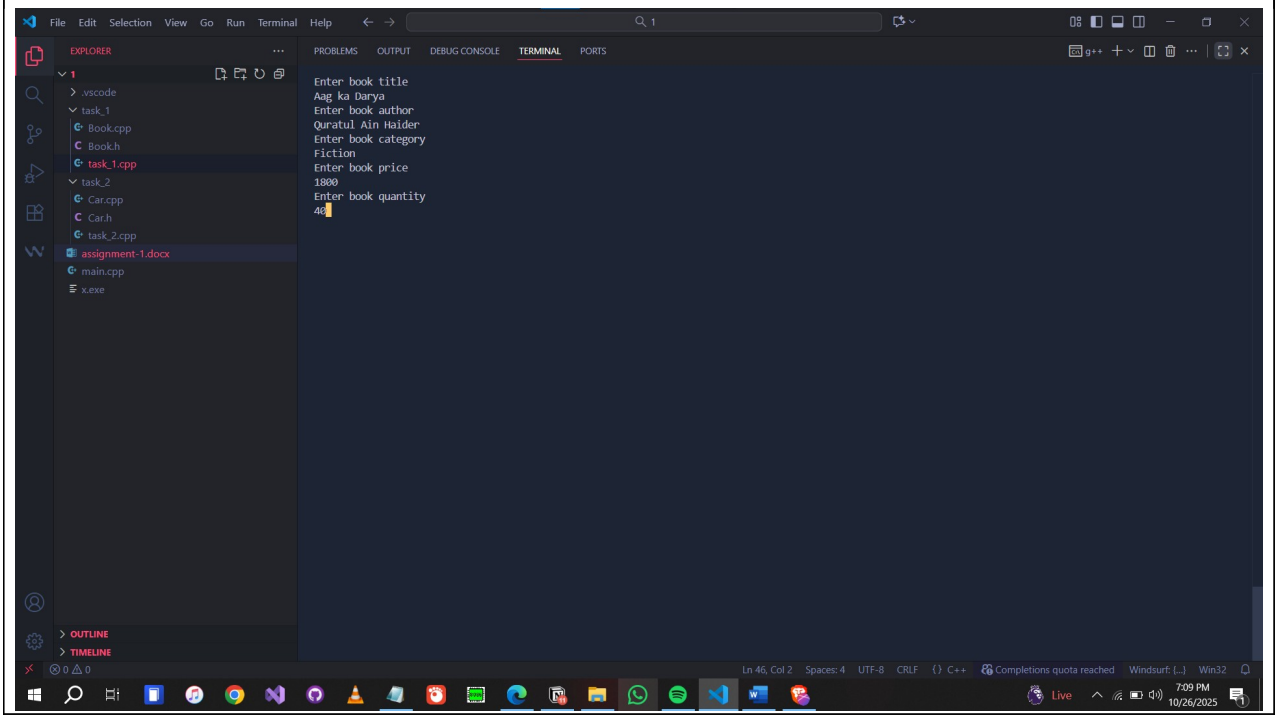
Void Book

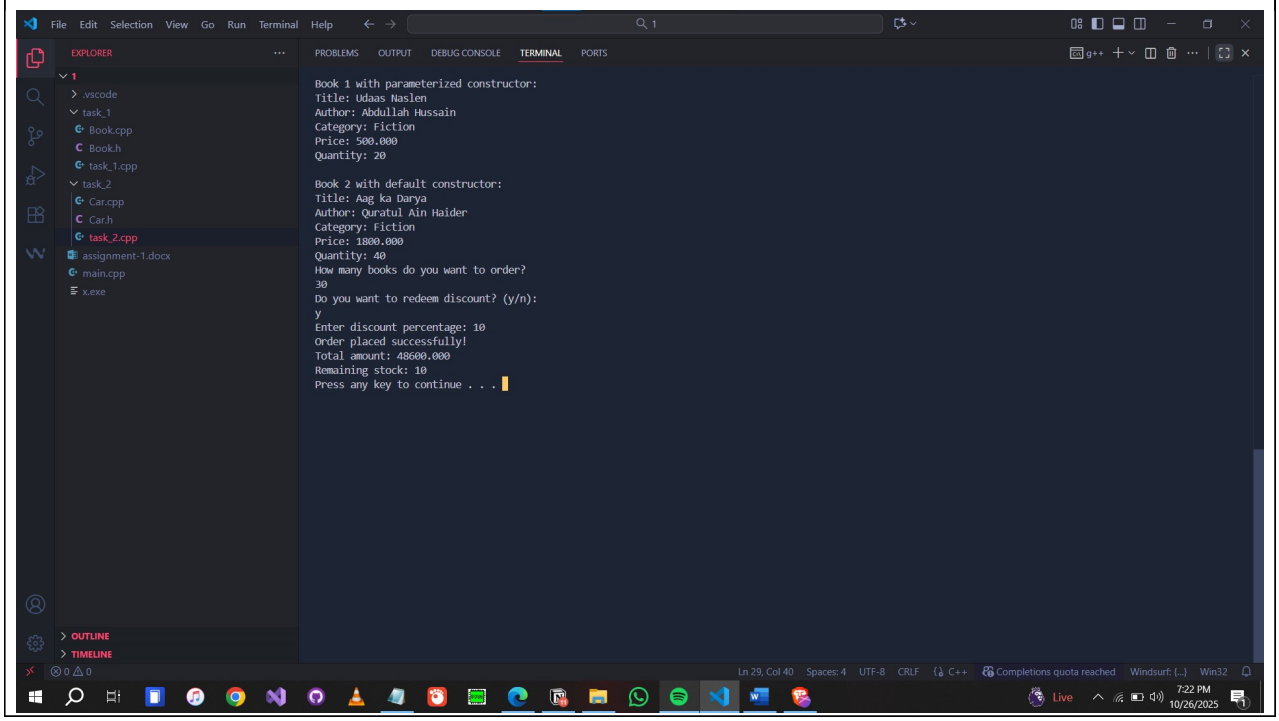
```
#include "Book.h"


void book() {
    system("cls");

    Book b_1("Udaas Naslen", "Abdullah Hussain", "Fiction", 500, 20);
    Book b_2;

    cout << "Enter book title" << endl;
    char* title = new char[20];
    cin.ignore();
    cin.getline(title, 20);
    b_2.setTitle(title);

    cout << "Enter book author" << endl;
    char* author = new char[20];
    cin.getline(author, 20);
    b_2.setAuthor(author);

    cout << "Enter book category" << endl;
    char* category = new char[20];
    cin.getline(category, 20);
    b_2.setCategory(category);
```

```
    cout << "Enter book price" << endl;
    double price;
    cin >> price;
    b_2.setPrice(price);

    cout << "Enter book quantity" << endl;
    int quantity;
    cin >> quantity;
    b_2.setQuantity(quantity);
    system("cls");
    cout << "Book 1 with parameterized constructor: " << endl;
    b_1.printBook();
    cout << endl << "Book 2 with updated values: " << endl;
    b_2.printBook();

    cout << "How many books do you want to order? " << endl;
    int n;
    cin >> n;
    b_2.orderBook(n);

    system("pause");
}
```

Screenshot:


File Edit Selection View Go Run Terminal Help
EXPLORER
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
*** Menu ***
1. Book
2. Car
0. Exit
Enter your choice (1 - 2): 1
.vscode
task_1
Book.cpp
Book.h
task_1.cpp
task_2
Car.cpp
Car.h
task_2.cpp
assignment-1.docx
main.cpp
x.exe
OUTLINE
TIMELINE
Ln 46, Col 2 Spaces: 4 UTF-8 CRLF C++ Completions quota reached Windsurf: (...) Win32
Live
7:07 PM
10/26/2025


File Edit Selection View Go Run Terminal Help
EXPLORER
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Enter book title
Aag ka Darya
Enter book author
Quratul Ain Haider
Enter book category
Fiction
Enter book price
1800
Enter book quantity
40
.vscode
task_1
Book.cpp
Book.h
task_1.cpp
task_2
Car.cpp
Car.h
task_2.cpp
assignment-1.docx
main.cpp
x.exe
OUTLINE
TIMELINE
Ln 46, Col 2 Spaces: 4 UTF-8 CRLF C++ Completions quota reached Windsurf: (...) Win32
Live
7:09 PM
10/26/2025


File Edit Selection View Go Run Terminal Help
EXPLORER
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
1
.vscode
task_1
Book.cpp
Book.h
task_1.cpp
task_2
Car.cpp
Car.h
task_2.cpp
assignment-1.docx
main.cpp
x.exe
OUTLINE
TIMELINE
Book 1 with parameterized constructor:
Title: Udaas Naslen
Author: Abdullah Hussain
Category: Fiction
Price: 500.000
Quantity: 20
Book 2 with default constructor:
Title: Aag ka Darya
Author: Quratul Ain Haider
Category: Fiction
Price: 1800.000
Quantity: 40
How many books do you want to order?
30
Do you want to redeem discount? (y/n):
y
Enter discount percentage: 10
Order placed successfully!
Total amount: 48600.000
Remaining stock: 10
Press any key to continue . . .
Ln 29, Col 40 Spaces: 4 UTF-8 CRLF C++ Completions quota reached Windsurf: {...} Win32
Live
7:22 PM
10/26/2025

Question #2:

Write a class Car that has:

- Data members: brand, model, and price.
- A **default constructor** that sets default values ("Unknown", "N/A", 0).
- Constructor Overloads take all values
- A **function** display() that shows all details.

Task:

1. Create two objects: one using the default constructor and another using a **parameterized constructor**.
2. Display both cars' details.
3. Use getter and setter function to set and get the car attributes values

Code:

Car.h

```
#include <iostream>
#include <cstring>
#include <iomanip>
using namespace std;
class Car {
private:
   char* m_brand;
   char* m_model;
   int m_year;
   double m_price;
public:
```

```
    Car();
    Car(const char*, const char*, int, double);
    ~Car();

    void setValue(char*, char*, int, double);
    char* getBrand();
    char* getModel();
    int getYear();
    double getPrice();
    void display();
};
```

Car.cpp

```
#include "Car.h"

Car::Car() {
    m_brand = new char[strlen("Unknown") + 1];
    strcpy(m_brand, "Unknown");

    m_model = new char[strlen("N/A") + 1];
    strcpy(m_model, "N/A");

    m_year = 0;
    m_price = 0.0;
}

Car::Car(const char* brand, const char* model, int year, double price) {
    m_brand = new char[strlen(brand) + 1];
    strcpy(m_brand, brand);

    m_model = new char[strlen(model) + 1];
    strcpy(m_model, model);

    m_year = year;
    m_price = price;
}
```

```
Car::~Car() {
    delete[] m_brand;
    delete[] m_model;
}

void Car::setValue(char* brand, char* model, int year, double price) {
    delete[] m_brand;
    m_brand = new char[strlen(brand) + 1];
    strcpy(m_brand, brand);

    delete[] m_model;
    m_model = new char[strlen(model) + 1];
    strcpy(m_model, model);

    m_year = year;
    m_price = price;
}

char* Car::getBrand() {
    if (m_brand == nullptr) return nullptr;
    char* temp = new char[strlen(m_brand) + 1];
    strcpy(temp, m_brand);
    return temp;
```

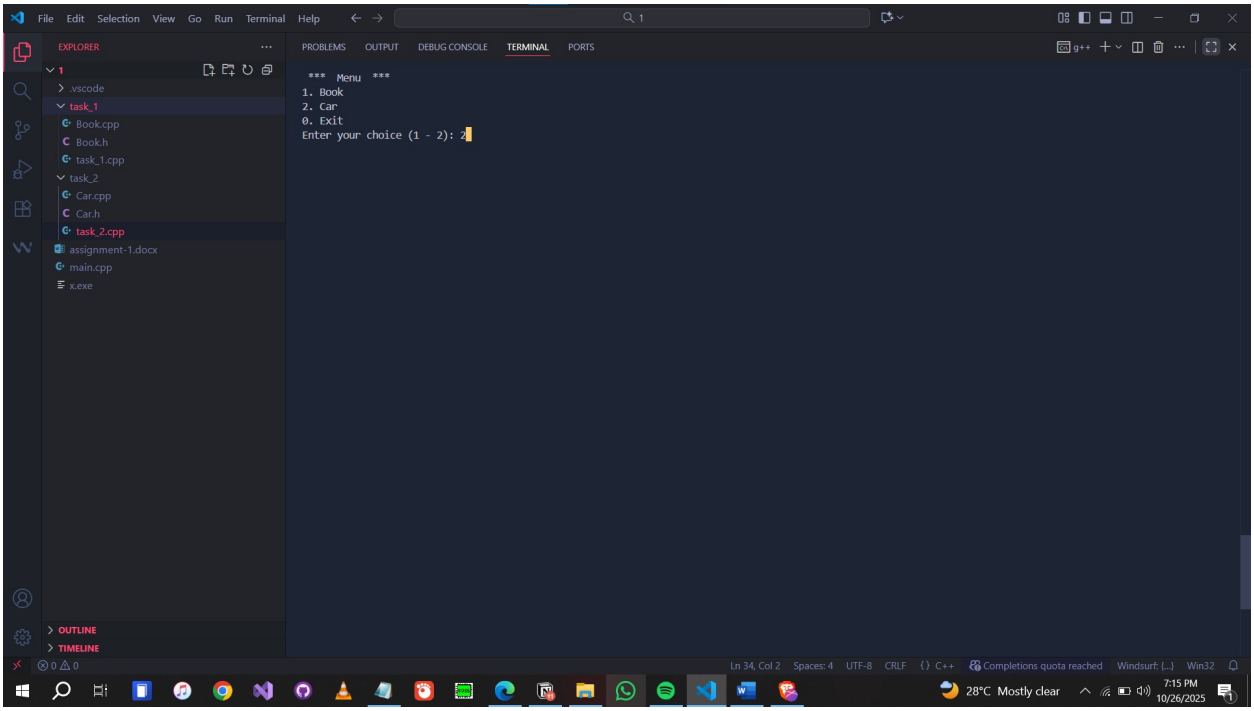
```
}

char* Car::getModel() {
    if (m_model == nullptr) return nullptr;
    char* temp = new char[strlen(m_model) + 1];
    strcpy(temp, m_model);
    return temp;
}

int Car::getYear() {
    return m_year;
}

double Car::getPrice() {
    return m_price;
}

void Car::display() {
     //This function can also get values without getters
    cout << "Brand: " << getBrand() << endl;
    cout << "Model: " << getModel() << endl;
    cout << "Year: " << getYear() << endl;
    cout << "Price: Rs. " << getPrice() << endl;
}
```

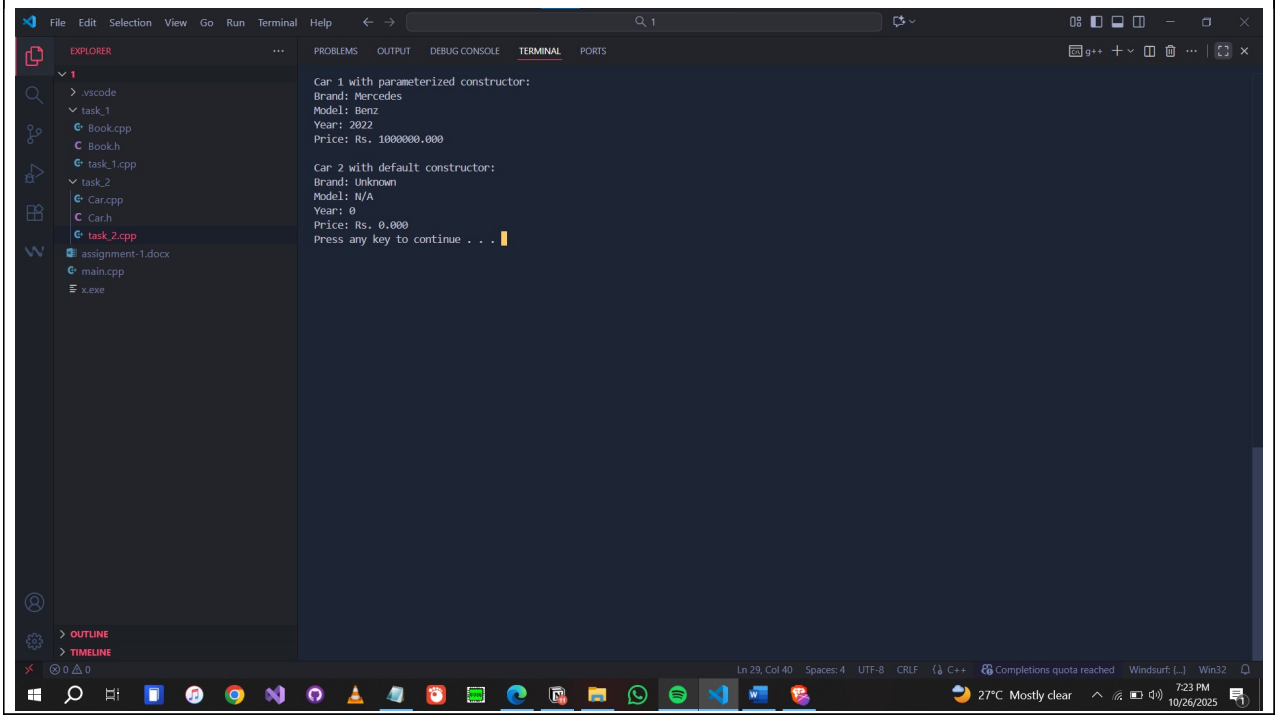
Void Car

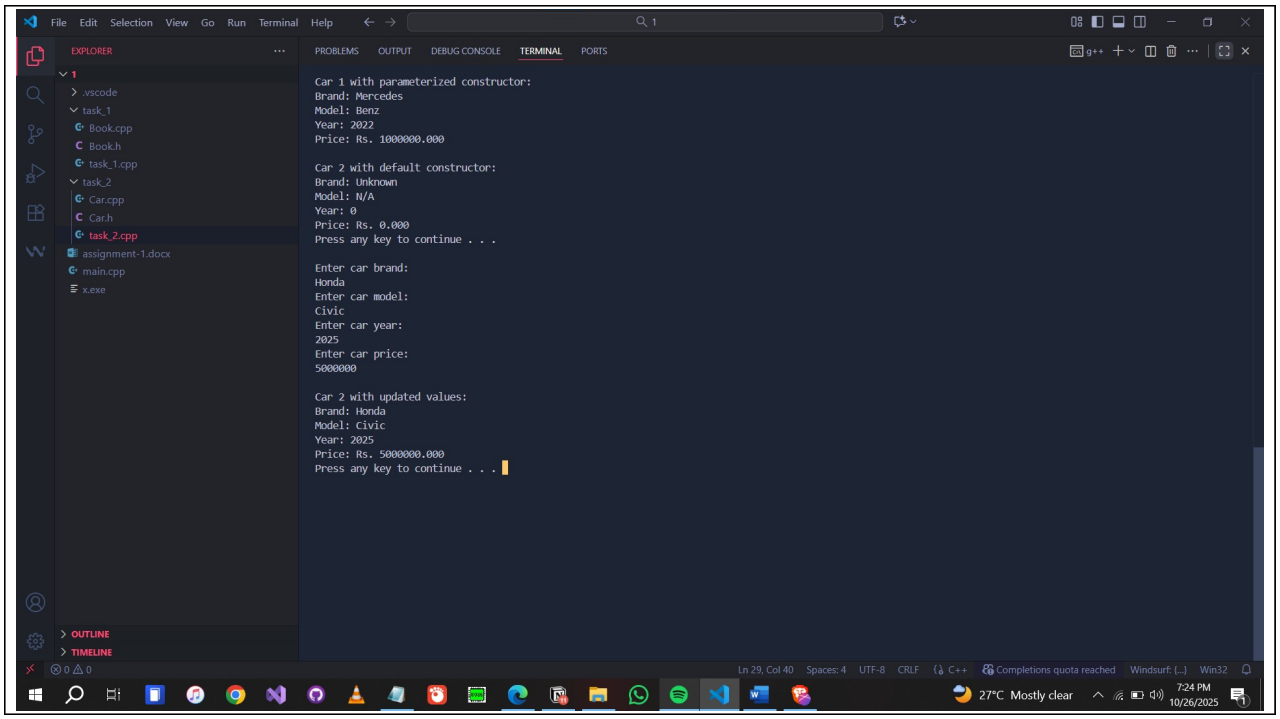
```
#include "Car.h"


void car() {
    system("cls");
    Car c_1("Mercedes", "Benz", 2022, 1000000);
    Car c_2;

    cout << "Car 1 with parameterized constructor: " << endl;
    c_1.display();
    cout << endl << "Car 2 with default constructor: " << endl;
    c_2.display();
    system("pause");

    char* brand = new char[20];
    char* model = new char[20];
    int year = 0;
    double price = 0;

    cout << endl << "Enter car brand: " << endl;
    cin.ignore();
    cin.getline(brand, 20);
    cout << "Enter car model: " << endl;
```

```
    cin.getline(model, 20);
    cout << "Enter car year: " << endl;
    cin >> year;
    cout << "Enter car price: " << endl;
    cin >> price;

    c_2.setValue(brand, model, year, price);

    cout << endl << "Car 2 with updated values: " << endl;
    c_2.display();

    system("pause");
}
```

Screenshot:


File Edit Selection View Go Run Terminal Help
EXPLORER
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
*** Menu ***
1. Book
2. Car
0. Exit
Enter your choice (1 - 2): 2
.vscode
task_1
Book.cpp
Book.h
task_1.cpp
task_2
Car.cpp
Car.h
task_2.cpp
assignment-1.docx
main.cpp
x.exe
OUTLINE
TIMELINE
28°C Mostly clear
7:15 PM
10/26/2025


File Edit Selection View Go Run Terminal Help
EXPLORER
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Car 1 with parameterized constructor:
Brand: Mercedes
Model: Benz
Year: 2022
Price: Rs. 1000000.000
Car 2 with default constructor:
Brand: Unknown
Model: N/A
Year: 0
Price: Rs. 0.000
Press any key to continue . . .
.vscode
task_1
Book.cpp
Book.h
task_1.cpp
task_2
Car.cpp
Car.h
task_2.cpp
assignment-1.docx
main.cpp
x.exe
OUTLINE
TIMELINE
27°C Mostly clear
7:23 PM
10/26/2025


File Edit Selection View Go Run Terminal Help
EXPLORER
1
.vscode
task_1
Book.cpp
Book.h
task_1.cpp
task_2
Car.cpp
Car.h
task_2.cpp
assignment-1.docx
main.cpp
x.exe
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Car 1 with parameterized constructor:
Brand: Mercedes
Model: Benz
Year: 2022
Price: Rs. 1000000.000
Car 2 with default constructor:
Brand: Unknown
Model: N/A
Year: 0
Price: Rs. 0.000
Press any key to continue . . .
Enter car brand:
Honda
Enter car model:
Civic
Enter car year:
2025
Enter car price:
5000000
Car 2 with updated values:
Brand: Honda
Model: Civic
Year: 2025
Price: Rs. 5000000.000
Press any key to continue . . .
Ln 29, Col 40 Spaces: 4 UTF-8 CRLF C++ Completions quota reached Windsurf: {...} Win32
27°C Mostly clear
7:24 PM
10/26/2025